

WELL COMPLETION REPORT (SYNTHETIC)

OIL-KMC-004 - Kumchai-004

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000048
Field / District	Kumchai / Tinsukia
Block	OALP-IV-04
Basin	Assam-Arakan
Formation	Barail (BAR)
Spud / Completion	2019-01-09 / 2019-05-20
Measured / TVD	2663 m / 2648 m
Trajectory	Deviated; inc 22.6 deg; az 131.5 deg
Formation Interval	1675-2623 m
Lithology	Sandstone-shale-coal sequence
Pore Pressure Reference	3399 psi
Fracture Gradient	15.57 ppg
Mud System	PHPA Water-Based Mud
NWIS Well Risk	Moderate (54.5/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	665	Surface
9 5/8 in casing	1544	Intermediate
7 in liner/casing	2343	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2019-02-14 868 m	Mud Loss	High	Partial returns observed at shale shaker. Mitigation: Stage circulation and monitor pit volume before resuming drilling. Lesson: Track flow balance and pit volume trend before losses become severe.
2019-02-19 976 m	Stuck Pipe	High	High drag was recorded while circulating remained possible. Mitigation: Apply approved freeing procedure and avoid excessive overpull. Lesson: Rising torque plus falling ROP should trigger hole-cleaning action.
2019-03-18 1491 m	Fishing Operation	High	Normal pipe-recovery attempts were unsuccessful. Mitigation: Review BHA failure mode before resuming drilling. Lesson: Maintain detailed BHA inventory for rapid recovery planning.
2019-04-07 1866 m	Fishing Operation	High	Normal pipe-recovery attempts were unsuccessful. Mitigation: Use approved fishing program with controlled overpull and circulation. Lesson: Maintain detailed BHA inventory for rapid recovery planning.
2019-04-20 2112 m	Mud Loss	High	Flow-out dropped below flow-in with pit loss. Mitigation: Optimize hydraulics and maintain conservative ROP through loss interval. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
2019-05-08 2433 m	Stuck Pipe	High	Hookload became erratic during pipe movement. Mitigation: Apply approved freeing procedure and avoid excessive overpull. Lesson: Avoid prolonged static periods in high-drag interval.
2019-01-23 406 m	Cementing Issue	Medium	Observed returns did not match calculated displacement. Mitigation: Use staged cementing response when returns deteriorate. Lesson: Volume imbalance is an important cement-placement warning.

Completion Summary

The synthetic well reached 2663 m MD in the Barail interval. The NWIS knowledge record links 15 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.